02452 Machine Learning - Project 1

Data: Feature extraction, and visualization

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GitHub Repository:
https://github.com/krusand/02452-ML-Project

Section/Subsection	Main Responsible
Introduction	Andreas
Data Summary	Andreas
Exploratory Data Analysis	Andreas
PCA Analysis	Sebastian
Discussion	Sebastian

Table 1: Responsible group member for each report section.

MSc, Human Centered AI
Technical University of Denmark

October 2025

1 Introduction to Dataset

Understanding the human health has been a crucial research topic throughout history. In the 1970s, it was observed that suspiciously many Afrikaans-speaking people suffered from ischaemic heart disease in South Africa. To understand the prevalence and intensity of the related risk factors, Rossouw et al. recruited more than 7000 subjects from the population of interest and recorded for each subject whether he or she suffered from coronary heart disease (CHD) along with a variety of known risk factors for heart diseases. A few years later, in 1987, Hastie and Tibshirani extracted a subset of 465 subjects from the original white male subjects as this group had a mortality rate two and a half times larger than the rate of the females. [1].
Our dataset contains the recorded data from the subset of male subjects. However, the recorded data for 3 subjects is missing, why the dataset contains 462 rows, representing 160 cases who did suffer from CHD and 302 control cases [1]. The dataset can be found on the book website *The Elements of Statistical Learning* [2].

1.1 Related work

We found two previous studies using this data, of which we focus on the study *Non-parametric logistic and proportional odds regression* [3]. This study focuses on the classification of coronary heart disease, by using the variables Systolic blood pressure, Cumulative tobacco intake, Cholesterol ratio, Psycho social stress (Type A), Age, Total energy expenditure in leisure time and occupational activities, and family history of CHD. They used a logistic regression model, and analyzed the covariates using Analysis of deviance (ANODEV). ANODEV showed that excluding the variables Type A, and Family history of CHD, raised the deviance the most. This means that Type A and family history of CHD were the most important for predicting CHD in the linear model. At first, the authors found that systolic blood pressure was associated with CHD for low and high values of systolic blood pressure, whereas the middle values were not. However, the authors then found that including whether patients/controls took medication for high systolic blood pressure, as an interaction term, showed that an increase in systolic blood pressure meant increased prevalence of CHD, as expected. The authors also showed that cumulative tobacco intake had a non-linear relationship with prevalence of CHD.

1.2 Our expectations

The problem of classification and regression is, in our case, two different problems. For classification, we want to predict `chd`. Since [3] showed which covariates were important, we wish to confirm their findings with our own analysis and expand using other variables. This would give insight into which covariates in the data are linked to `chd`. We consider the classification task our main machine learning aim.

For regression, we want to predict `obesity` (BMI). We want to learn whether covariates such as `alcohol`, `tobacco`, and `chd` can predict the BMI of patients/controls. We should consider transforming some variables, particularly one interesting variable found by the study [3] was cumulative tobacco intake, which looked like an exponential relationship. This means a log transform could be preferred.

2 Data Summary

The dataset consists of 10 variables. These 10 variables are shown in Table 2. Eight of these variables are continuous, ratio. Two variables are discrete, both nominal. There are no missing data, and to our knowledge no data problems.

In Table 3, we show some summary statistics of the variables in the dataset. The average age of participants is about 43 years old, with participants ranging between 15 and 64 years old. A bit more than 50% of these are overweight, with a BMI over 25. The tobacco use is right skewed, meaning that the mean is higher than the median. This indicates there are few participants who use a substantial amount of tobacco, and most use very little. This is also indicated by the maximum amount of tobacco (31 kg!). Alcohol consumption shows the same tendency.

Table 2: Variable descriptions.

Variable	Description
sbp (cont. ratio)	Systolic blood pressure in millimetres of mercury (mm Hg).
tobacco (cont. ratio)	Cumulative tobacco use in kilograms.
ldl (cont. ratio)	Low density lipoprotein cholesterol.
adiposity (cont. ratio)	Measurement of obesity. We will probably not use this and instead use the `obesity` variable.
famhist (disc. nominal)	previous family history of ischaemic heart disease (0: absent, 1: present)
typea (cont. ratio)	Type-A coronary prone personality behaviour as measured by a self-administered Bortner Short Rating Scale. Possible total scores can range from 12 to 84. Rossouw et al. (1983) "arbitrarily" classify those with scores of 55 or more "as exhibiting type A behaviour."
obesity (cont. ratio)	A measure of obesity; body mass index (or BMI).
alcohol (cont. ratio)	Current alcohol consumption. Unit is uncertain, but higher means more alcohol.
age (cont. ratio)	Age in years at time of study.
chd (disc. nominal)	The response, a factor identifying whether the subject had been diagnosed as having coronary heart disease or not.

3 Visualizing the Dataset

3.1 Exploratory Data Analysis

In Figure 1, we have plotted histograms of each attribute.

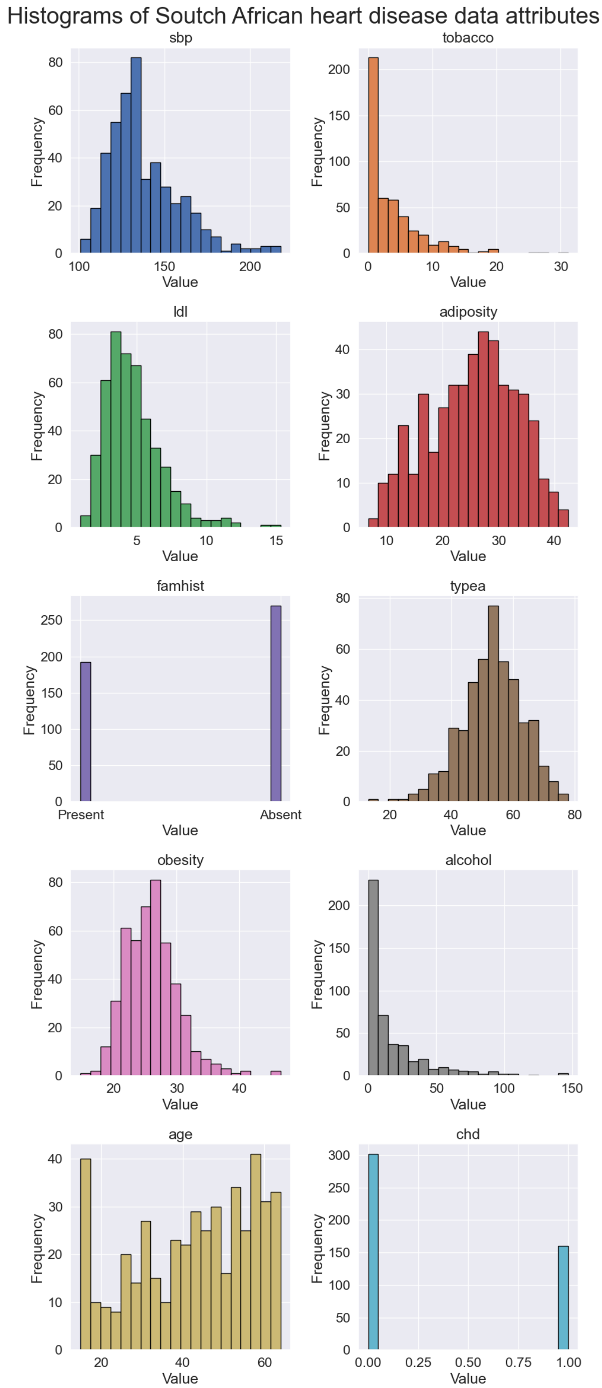

Figure 1: Histogram for each of the 10 variables.

From this, we see some attributes which are normally distributed, or at least look like they are normally distributed, particularly `typea`, `adiposity`

Table 3: Summary statistics of the South African Heart Disease dataset.

Statistic	sbp	tobacco	ldl	adiposity	typea	obesity	alcohol	age	famhist	chd
mean	138.33	**3.64**	4.74	25.41	53.10	26.04	17.04	**42.82**	0.42	0.35
std	20.50	4.59	2.07	7.78	9.82	4.21	24.48	14.61	0.50	0.48
min	101.00	0.00	0.98	6.74	13.00	14.70	0.00	**15.00**	0.00	0.00
25%	124.00	0.05	3.28	19.78	47.00	22.99	0.51	31.00	0.00	0.00
50%	134.00	**2.00**	4.34	26.12	53.00	**25.81**	7.51	45.00	0.00	0.00
75%	148.00	5.50	5.79	31.23	60.00	28.50	23.89	55.00	1.00	1.00
max	218.00	**31.20**	15.33	42.49	78.00	46.58	147.19	**64.00**	1.00	1.00

and `obesity`. The attributes `tobacco` and `alcohol` seem to be distributed by an exponential distribution. Since both `famhist` and `chd` are binary, both are Bernoulli distributed. Both `sbp` and `ldl` look like a chi-squared distribution while age looks like it is distributed with more weight towards older ages. Both `alcohol` and `tobacco` have some values which look extreme, about 150 for alcohol, and 30 for tobacco, but given the exponentially distributed data, we would expect it since exponentially distributed data has a long tail.

To inspect the correlation of the different attributes, we plot a correlation matrix. This can be seen in Figure 2. The correlation matrix shows that `obesity` and `adiposity` are positively correlated. This is not surprising since both are measures of obesity. For our analysis, we should consider only having one of these attributes. Age is positively correlated with both `tobacco` and `adiposity`. The positive correlation with `tobacco` makes sense as tobacco is cumulative intake. When you get older, you have simply had the opportunity to intake more tobacco compared to younger people. Of course, tobacco is probably also more popular with older people. The correlation of age with `adiposity` also makes sense since people tend to move less and eat more as they age. It is important to look at which variables are correlated with chd. For the classification, this is what we want to predict. From the correlation matrix, it looks like the best contenders are age, `tobacco`, `famhist`, `ldl` and `adiposity`. Type A looks to not be correlated with chd, in contrast to what the previous study mentioned in Related work.

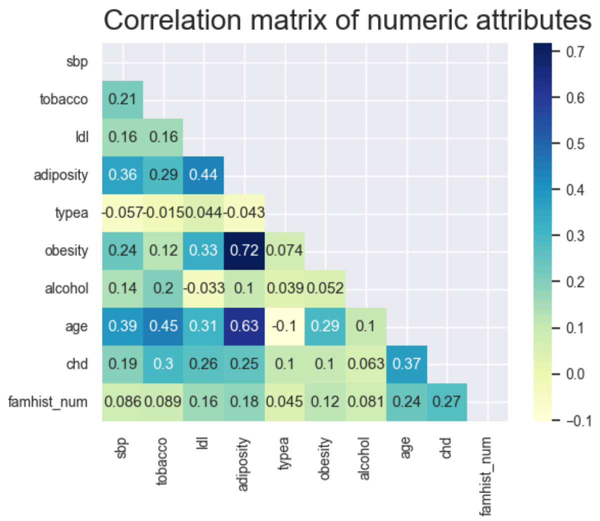

Figure 2: Correlation matrix.

To get an idea of how much overlap there is when conditioning on chd, we plot a pairplot. We show this in Figure 3. From the densities on the main diagonal, we see there is a majority of overlap between those with and without chd. Some attributes, like age, `adiposity`, `tobacco`, and `typea` show the positive (chd=1) density to be more skewed compared to the negative (chd=0). Once again, it looks like outliers may not be a big problem.

For all the scatterplots in Figure 3, the points of positive and negative chd look scattered on top of each other. This suggests there is not one feature alone which can predict chd, but multiple features are needed to separate the two groups. To look at all the data in one plot, we perform principal component analysis.

3.2 PCA Analysis

Since the heart disease dataset contains multiple variables, as shown in Table 2, it would require an abstract high-dimensional space to model the entire dataset. Therefore, it is relevant to

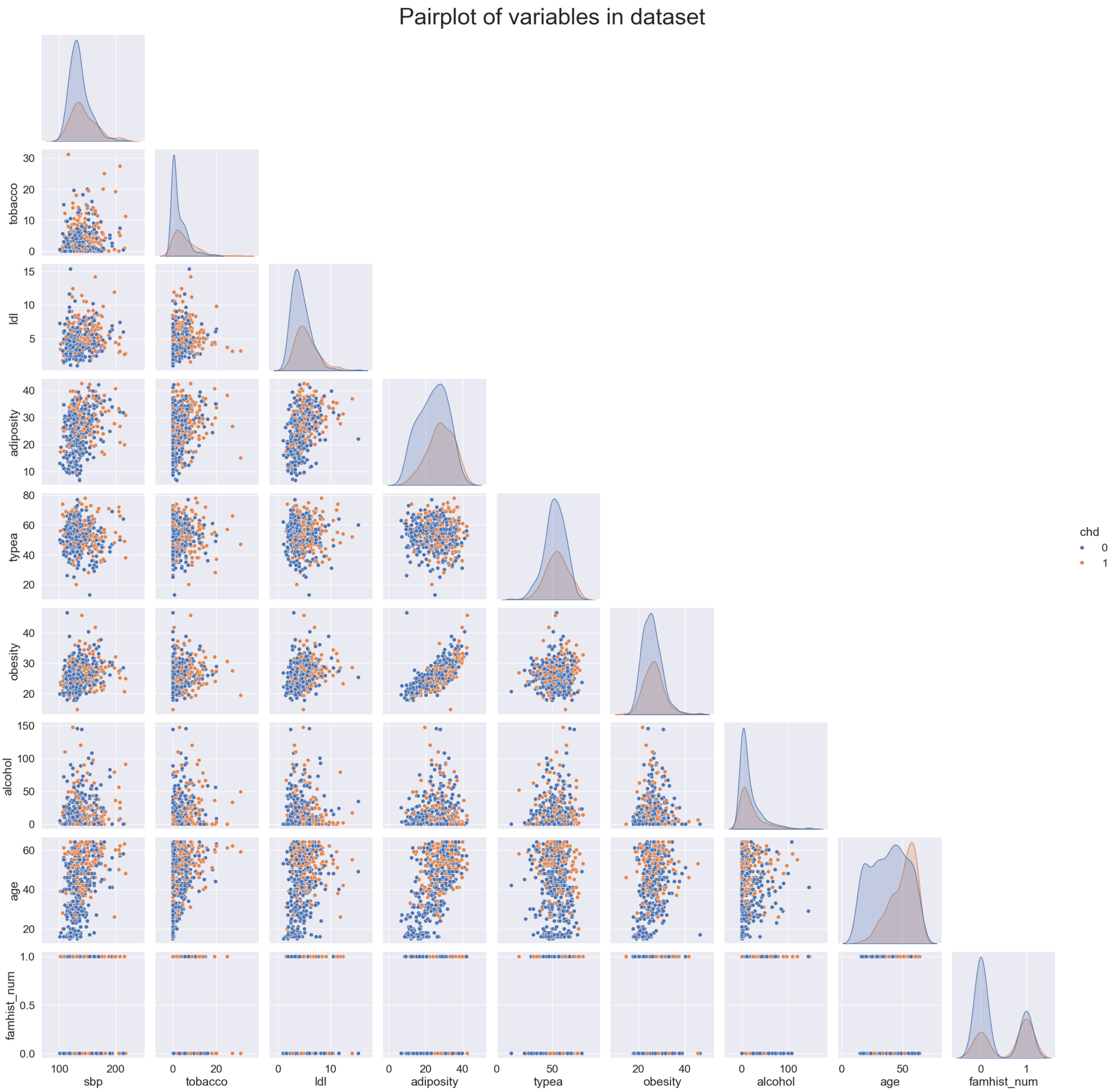

Figure 3: Pairplot between each of the eight continuous variables.

investigate whether the data can be projected onto a lower-dimensional subspace that is capable of explaining the variation within the data. This technique is known as principal component analysis (PCA) and will be carried out in this subsection.

We include all continuous variables in our PCA analysis, resulting in eight possible principal components. The variables `famhist` and `chd` are excluded from the PCA analysis. Before we compute the principal components, we standardize the data to ensure that the scale of each variable does not affect the magnitude of the corresponding coefficients for each component.

We obtain principal components, using an instance of the `PCA` class from `sklearn.decomposition`, fitted onto the standardized data matrix, $\tilde{X}$. This performs the singular value decomposition, $\tilde{X} = U\Sigma V^T$, where the columns of V are the principal components indicating how much and in which direction each variable contributes to each component. We obtain the components by applying the `components_` method onto the fitted PCA object and then transposing.

For our components, most variables have a positive impact on the first component, in particular `age` and `adiposity`. For the second component, `alcohol` and `tobacco` have the largest positive impact, while `ldl`, `obesity`, and `typea` have a negative impact. The impacts for each of the two first components are represented by the vectors within a unit circle in Figure 4.

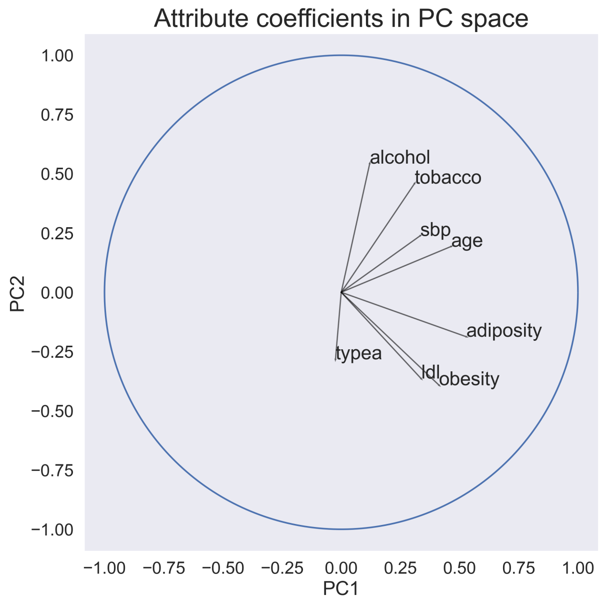

Figure 4: Direction of loading vectors for each variable using the two first components.

The coefficients can also be shown as bars to include multiple components while still plotting in two dimensions. This is shown in Figure 5 for the first three components. The bars representing PC1 and PC2 align with the vectors shown in Figure 4.

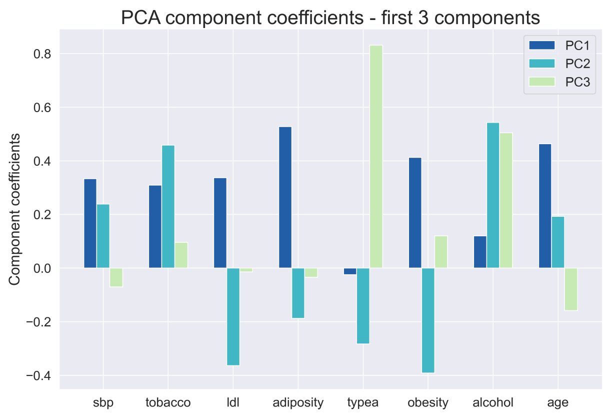

Figure 5: Coefficients of each variable for the first three principal components.

The first component has positive loadings for most variables and seems to represent general health factors such as `age`, `obesity`, `sbp`, and `tobacco`, where a higher value implies worse health.

PC2 distinguishes people with high `ldl`, `obesity`, and `typea` values from people with high `alcohol` and `tobacco` consumption. An interpretation could be that PC2 distinguishes between physical and behavioral health factors.

The third component has a strong positive loading for the `typea` variable, which signifies psychological traits such as impatience and stress [4]. Thus, PC3 primarily distinguishes between people based on their self-reported type A score.

Only three out of eight principal components have been assessed in Figure 4 and 5. But how many components do we need to represent the data? The components explain a decreasing proportion of the variation within the data based on their corresponding eigenvalue. Figure 6 shows the proportion of explained variance for each component individually and accumulated.

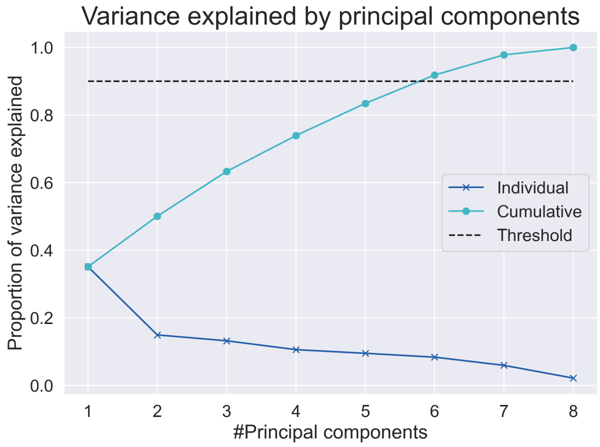

Figure 6: Individual and accumulated proportion of explained variance for a given number of principal components.

A common threshold for the proportion of explained variance is 90%, which in our case is achieved when using six out of the eight components. Thus, it is possible to project the data onto a slightly lower-dimensional subspace and still be able to explain a majority of the variation within the data. It is also notable that the first component, PC1, explains almost 40% of the variation, compared to less than 20% for each of the remaining components.

As the final part of the PCA analysis, the standardized data, $\tilde{X}$ will be projected onto the PCA space spanned by the first two principal components. In this way, it is possible to visualize the high-dimensional data in a two-dimensional scatterplot with PC1 and PC2 as the new axes. Such a plot is shown in Figure 7, where each projected data point is color-coded according to the `chd` variable.

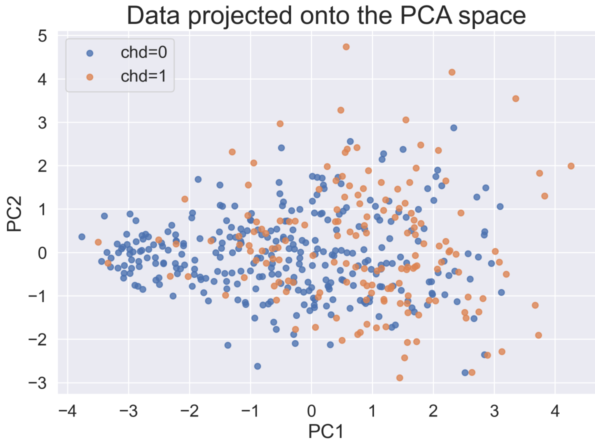

Figure 7: Standardized data projected onto the PCA space spanned by PC1 and PC2.

In Figure 7 it is hard to distinguish the classes `chd=0` and `chd=1` from each other within the space spanned by PC1 and PC2. This can be explained by the fact that the first two components only explain 50% of the total variation within the data. In other words, more components are probably needed in order to separate the two classes properly.

4 Discussion of Results

To summarize our findings, the data summary showed that the South African Heart Disease dataset represents a large variety of individuals spanning across old and young people as well as underweight and overweight people. The summary together with the histograms in Figure 1 also reveals that alcohol and tobacco consumption is right-skewed with most people having a low consumption and a long tail of a few people with high consumptions.

From the exploratory data analysis, we learned that the `adiposity` and `obesity` variables are strongly positively correlated, which is why we should only include one of those variables in our regression and classification models. Specifically for the classification task, `age`, `tobacco`, `famhist`, `ldl`, and `adiposity` seem like the most suitable variables to predict the class given by the `chd` variable. This is supported by the pairplot in Figure 3 which reveals that no single variable is able to clearly distinguish the two classes of `chd`.

The main take-away message from the PCA analysis is that by standardizing our dataset, it is possible to project the data onto a meaningful subspace of lower dimensionality. In this way, we can represent the data in fewer dimensions while still being able to explain a majority of the variability within the data.

Overall, we believe that we are able to make more informed modeling choices when conducting the classification task, which is our primary machine learning aim. Both in terms of which variables to choose, how to deal with them, and possibly how to use fewer dimensions to achieve our aim.

5 Use of Generative AI

We have consulted generative AI for guidance on how to format some of the plots, e.g., on how to adjust the font size. Furthermore, we used generative AI to format tables in LaTeX.

References

[1] R.W. Oldford. South African Heart Disease Data. Retrieved September 23, 2025 from https://great-northern-diver.github.io/loon.data/reference/SAheart.html?utm_source=chatgpt.com#author-1

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